

```
In [ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
%matplotlib inline
```

```
In [ ]: df = pd.read_csv("loan_data_set.csv")
```

```
In [ ]: df.head()
```

```
In [ ]: df.shape
```

```
In [ ]: df.info()
```

```
In [ ]: df.describe()
```

```
In [ ]: df.isna().sum()
```

```
In [ ]: df["ApplicantIncome"].plot(kind="hist")
plt.show()
```

```
In [ ]: df["ApplicantIncome"].fillna(df["ApplicantIncome"].mean(), inplace=True)
df
```

```
In [ ]: df["CoapplicantIncome"].plot(kind="hist")
plt.show()
```

```
In [ ]: df["CoapplicantIncome"].fillna(df["CoapplicantIncome"].mean(), inplace=True)
df
```

```
In [ ]: df["LoanAmount"].plot(kind="hist")
plt.show()
```

```
In [ ]: df["LoanAmount"].fillna(df["LoanAmount"].mean(), inplace=True)
```

```
In [ ]: df["Credit_History"].plot(kind="hist")
plt.show()
```

```
In [ ]: df["Credit_History"].fillna(np.random.randint(0,2), inplace=True)
df
```

```
In [ ]: grouped_df = df[["ApplicantIncome", "CoapplicantIncome", "LoanAmount", "C
```

```
In [ ]: mean = grouped_df.mean()
mean
```

```
In [ ]: median = grouped_df.median()
median
```

```
In [ ]: min = grouped_df.min()  
min
```

```
In [ ]: max = grouped_df.max()  
max
```

```
In [ ]: std = grouped_df.std()  
std
```

```
In [ ]: data = pd.read_csv("Iris.csv")  
data.keys()
```

```
In [ ]: data.shape
```

```
In [ ]: data.info()
```

```
In [ ]: data.describe()
```

```
In [ ]: data["SepalLengthCm"].describe()
```

```
In [ ]: data.groupby("Species").describe()
```

```
In [ ]: data.groupby("Species").describe().sum()
```

```
In [ ]: data.groupby(["Species"])[ "SepalLengthCm"].mean()
```

```
In [ ]: data.info()
```

```
In [ ]: data['Species'].value_counts()
```

```
In [ ]: data.Species.mode()
```

```
In [ ]: import seaborn as sns  
sns.set_style("whitegrid")  
sns.FacetGrid(data, hue="Species", height=7).map(plt.scatter, "SepalLengthCm", "PetalLengthCm")  
plt.show()
```

```
In [ ]: setosa_data = data[data['Species'] == 'Iris-setosa']  
versicolor_data = data[data['Species'] == 'Iris-versicolor']  
virginica_data = data[data['Species'] == 'Iris-virginica']
```

```
In [ ]: setosa_data
```

```
In [ ]: versicolor_data
```

```
In [ ]: virginica_data
```

```
In [ ]: # Setosa  
setosa_stats = {  
    'percentile': np.percentile(setosa_data['SepalLengthCm'], 50),  
    'mean': np.mean(setosa_data['SepalLengthCm']),  
    'std_dev': np.std(setosa_data['SepalLengthCm'])  
}
```

```
# Versicolor
versicolor_stats = {
    'percentile': np.percentile(versicolor_data['SepalLengthCm'], 50),
    'mean': np.mean(versicolor_data['SepalLengthCm']),
    'std_dev': np.std(versicolor_data['SepalLengthCm'])
}

# Virginica
virginica_stats = {
    'percentile': np.percentile(virginica_data['SepalLengthCm'], 50),
    'mean': np.mean(virginica_data['SepalLengthCm']),
    'std_dev': np.std(virginica_data['SepalLengthCm'])
}

# Print results
print("Statistics for Iris Setosa:")
print(setosa_stats)

print("\nStatistics for Iris Versicolor:")
print(versicolor_stats)

print("\nStatistics for Iris Virginica:")
print(virginica_stats)
```

In []:

In []: